GCN Analysis: Do Network Dependencies Explain Variance Beyond Individual SDoH in Study Enrollment?

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1. Problem and Dataset

This analysis asks whether geographic contiguity of ZIP codes explains additional variance in study enrollment beyond individual-level Social Determinants of Health (SDoH). Residents of adjacent ZIP codes share structural exposures, including proximity to clinics, transit access, and community resources, that shape research participation. Understanding which factors predict enrollment is important for identifying and addressing disparities in access to clinical research.

The dataset comes from the MS CATCH cohort study conducted at a large academic medical center in the San Francisco Bay Area. The full dataset contains 1,398 matched case-control records: 700 MS patients and 698 matched controls, matched on age, gender, race, ethnicity, and MS onset year. The outcome of interest is study enrollment (enrolled = 1 vs. enrolled = 0), which is available for 212 patients: 104 enrolled and 108 in the comparison group. The remaining 1,187 records have no enrollment label, as they fall outside the enrollment tracking window or are matched controls not subject to the enrollment process.

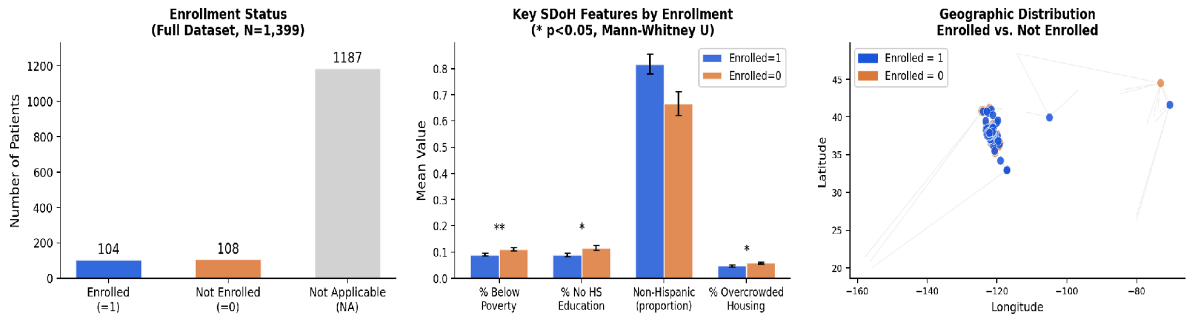
Each patient record is linked to a residential ZIP code, enabling integration of 17 area-level SDoH measures from the American Community Survey (ACS) and the Area Deprivation Index (ADI). These include poverty rate, educational attainment, median household income, housing overcrowding, neighborhood walkability, and a Charlson comorbidity score. Geographic coordinates are available for each ZIP code centroid, enabling construction of the spatial graph used by the GCN.

Table 1 summarizes key dataset characteristics. Among the 212 patients with enrollment labels, enrolled patients live in significantly less deprived neighborhoods (poverty rate 9.0% vs. 11.1%, p = 0.010), are more likely to be Non-Hispanic (81.7% vs. 66.7%, p = 0.013), and show lower rates of housing overcrowding (4.7% vs. 5.9%, p = 0.019). These patterns suggest socioeconomic and demographic gradients in enrollment access that motivate the use of a graph-based spatial model.

Table 1. Dataset Summary and Key Enrollment Comparisons

Variable	Enrolled=1(n=104)	Enrolled=0(n=108)	Note
Total (labeled)	104	108	1,187 NA excluded
Non-Hispanic (%)	81.7%	66.7%	p = 0.013 *
% Below Poverty	9.0%	11.1%	p = 0.010 **
% No HS Education	8.9%	11.6%	p = 0.010 *
% Overcrowded Hous.	4.7%	5.9%	p = 0.019 *
Comorbidity Score	0.24	0.42	p = 0.078 (trend)
Unique ZIP codes	~100	~90	San Francisco Bay Area
GCN graph nodes	212	patients	Patient-level graph
GCN graph edges	486	edges	Shared/adjacent ZIP

Figure 1. Dataset Description: Enrollment Status, Key SDoH Feature Comparisons, and Geographic Distribution



Left: enrollment status across the full dataset (N=1,399). Center: mean values of the four most discriminating SDoH features by enrollment group; error bars show standard error; asterisks indicate $p < 0.05$ (Mann-Whitney U). Right: geographic distribution of enrolled (blue) and comparison (orange) patients; node size proportional to patient count per ZIP; lines show Voronoi contiguity structure.

2. Algorithm: Graph Convolutional Network at Patient Level

A Graph Convolutional Network (GCN) is selected because enrollment is shaped by a patient's own neighborhood conditions and the conditions of surrounding neighborhoods. Patients living in geographically proximate ZIP codes share access to the same clinics, transportation infrastructure, and community outreach programs, all of which influence research participation. The GCN explicitly models these spatial dependencies by propagating SDoH information across edges in a patient-level graph. It is compared against logistic regression on individual SDoH features, which treats each patient independently.

The key assumption of the GCN is homophily: patients connected by an edge are assumed to share similar enrollment-relevant neighborhood characteristics. This is plausible because edges connect patients in the same or geographically contiguous ZIP codes. The GCN performs well when node features are informative, graph structure reflects meaningful relational proximity, and the neighborhood signal adds information beyond what individual features capture. All three conditions are at least partially met in this dataset, though the small labeled sample ($n = 212$) limits the strength of the signal the model can exploit.

The graph $G = (V, E)$ is defined at the patient level. Each node i in V represents one patient and carries a 17-dimensional SDoH feature vector x_i specific to that individual's residential ZIP code. Each edge (i, j) in E connects two patients who either share the same ZIP code or reside in Voronoi-contiguous ZIP codes, meaning their ZIP centroids are geographic neighbors with no other ZIP between them. The Voronoi tessellation partitions the plane so that every geographic point belongs to its nearest ZIP centroid; shared Voronoi cell boundaries define contiguity. The graph contains 212 nodes and 486 edges with a mean degree of 4.58; 13 patients have no neighbors and receive only their own features. Operating at the patient level preserves individual SDoH heterogeneity that would otherwise be lost by averaging features to the ZIP level.

The GCN uses fixed-kernel mean aggregation, holding weight matrices at identity rather than learning them from data. This is a deliberate design choice: with $n = 212$ labeled observations, learnable weight matrices would introduce more parameters than the data can reliably estimate, leading to overfitting (Wu et al., 2019). Each patient node is initialized with its own SDoH feature vector:

$$h_i^{(0)} = x_i$$

At each layer l, the node representation is updated by mean-aggregating over all patients connected by an edge, N(i):

$$h_i^{(l)} = MEAN\{\ h_j^{(l-1)}\ :\ j \in N(i) \cup \{i\}\ \}$$

In matrix form, self-loops are added via A~ = A + I so each node includes itself in aggregation. The degree matrix D~ = Σ A~ normalizes by neighborhood size. The GCN propagation rule at layer l is:

$$H^{(l)} = D\sim^{-1} A\sim H^{(l-1)}, \quad A\sim = A + I$$

With W^(l) = I, the propagation simplifies to repeated application of the normalized graph shift operator S = D~^(-1) A~, equivalent to iterated Laplacian smoothing:

$$H^{(l)} = S^{l} X, \quad S = D\sim^{-1} A\sim$$

Two layers are applied (l = 1, 2), so each patient incorporates SDoH information from patients up to two graph hops away. The final per-patient GCN feature vector concatenates representations from all layers, producing a 60-dimensional vector that encodes local, one-hop, and two-hop neighborhood context:

$$z_i = CONCAT(h_i^{(0)}, h_i^{(1)}, h_i^{(2)}) \in R^{3d}$$

This feature vector is passed to logistic regression, which predicts enrollment probability. For the baseline model, the 20-dimensional individual SDoH feature vector x_i is used directly. Both models use the same logistic regression classifier:

$$P(y = 1 \mid z_i) = \sigma(w^T z_i + b), \quad \sigma(t) = \frac{1}{1+e^{-t}}$$

Parameters are estimated by minimizing binary cross-entropy over the n = 212 labeled patients:

$$L = -\frac{1}{N} \Sigma_i \left[y_i \log \sigma(w^T z_i) + (1 - y_i) \log(1 - \sigma(w^T z_i)) \right]$$

Performance is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUROC) under 5-fold stratified cross-validation, which accounts for the near-balanced class distribution (104 vs. 108) and the small sample size.

3. Results and Discussion

Table 2 summarizes AUROC performance for both models under 5-fold cross-validation. Logistic regression on individual SDoH features (AUROC = 0.565) marginally outperforms the patient-level GCN (AUROC = 0.554). Both models achieve modest discrimination above chance (AUROC = 0.500), consistent with the small labeled sample (n = 212) and the inherently multi-factorial nature of research enrollment decisions.

Table 2. Model Performance Comparison (5-Fold Cross-Validation)

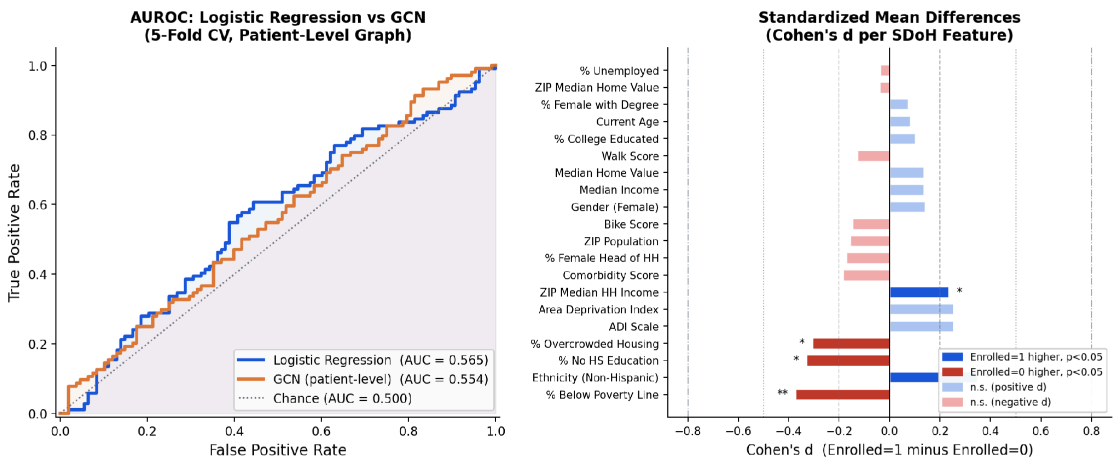
Model	AUROC	Description
Logistic Regression	0.565	Individual SDoH features; no graph structure
GCN (patient-level)	0.554	Fixed-kernel GCN; Voronoi patient graph; 2 layers
Chance baseline	0.500	Random classifier reference

The result that individual SDoH features marginally outperform the GCN is informative. It suggests that patient-level spatial proximity, as defined by shared or Voronoi-contiguous ZIP codes, does not capture enrollment-relevant neighborhood structure beyond what individual deprivation measures already encode under a linear model. The GCN aggregates the same SDoH variables across the graph, so when individual features are already informative, the smoothed neighborhood representations add limited additional signal.

Figure 2 presents the AUROC curves (left panel) and Cohen's d effect sizes for all 20 SDoH features (right panel). Five features reach statistical significance ($p < 0.05$) with small-to-medium effect sizes ($d = 0.24$-0.37): percentage below the poverty line ($d = -0.37$), Non-Hispanic ethnicity ($d = +0.35$), percentage with no high school education (d = -0.33), percentage in overcrowded housing (d = -0.31), and ZIP-level median household income (d = +0.24). Negative d values indicate the feature is higher among the comparison group, reflecting that enrolled patients live in less socioeconomically deprived neighborhoods. Age, walkability, and most housing value measures show effect sizes near zero, confirming that the enrollment gap is driven primarily by socioeconomic and demographic factors rather than overall neighborhood quality.

Both models perform modestly, which reflects genuine constraints of the data rather than model failure. With 212 labeled observations, 153 distinct ZIP codes, and complex multi-factorial determinants of enrollment, neither a graph-based nor a purely individual model is expected to achieve high discrimination. The GCN's fixed-kernel implementation is a principled response to this constraint, but it also means the model cannot learn which neighborhood features are most relevant to enrollment. A fully trained GCN with learnable aggregation weights, ReLU activations between layers, and appropriate regularization would be more expressive, but would require a substantially larger labeled cohort to avoid overfitting. Future work should expand enrollment tracking across the full matched dataset, apply a measure of spatial autocorrelation to formally test for spatial autocorrelation in enrollment before deploying graph-based methods, and investigate whether the enrollment disparities identified here persist after adjusting for clinical variables such as disease severity and time since diagnosis.

Figure 2. Results: AUROC Comparison and Cohen's d Effect Sizes by SDoH Feature



Left: AUROC curves for logistic regression and patient-level GCN under 5-fold cross-validation; shaded regions show area under each curve. Right: Cohen's d for all 20 SDoH features (enrolled=1 minus enrolled=0); dark blue bars indicate enrolled=1 is higher ($p < 0.05$), dark red bars indicate enrolled=0 is higher ($p < 0.05$), light bars are not significant; dashed vertical lines mark small (0.2), medium (0.5), and large (0.8) effect size thresholds.

References

Kipf, T. N. and Welling, M. (2017). Semi-supervised classification with graph convolutional networks. ICLR.

Wu, F., Souza, A., Zhang, T., Fifty, C., Yu, T., and Weinberger, K. (2019). Simplifying graph convolutional networks. ICML.